Automated Data Mining and Pattern Discovery System for Unstructured News Corpora

Section 1 — Introduction

The exponential growth of digital news necessitates a move beyond traditional keyword retrieval toward an **Automated Data Mining and Pattern Discovery System**. Unlike conventional cloud-dependent engines, this project architects a standalone, **Offline-First hybrid AI system** designed to transform unstructured corpora into mathematically backed transparent intelligence. The engineering baseline is built on an **embedded serverless DuckDB instance** (news_corpus.duckdb), enabling high-speed local processing of heterogeneous text without cloud latency. To ensure "Explainable AI" (XAI), the system integrates **Small Language Models via Ollama and LM Studio**, acting strictly as analytical interpreters to explain native corpus patterns without hallucination.To provide a high-level overview of the end-to-end system design, the core operational flow is illustrated below:

```
graph TD
    A[News Dataset / Hugging Face] --> B[Data Cleaning & Normalization]
    B --> C[Feature Extraction TF-IDF]
    C --> D[Data Mining Algorithms]
    D --> E[Pattern Discovery / Clustering / Classification]
    E --> F[Analytics Dashboard / Tableau]

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    classDef highlight fill:#d4ebf2,stroke:#005b96,stroke-width:2px;
    class A,F highlight;
```

Figure 1: High-level system architecture demonstrating the end-to-end data mining pipeline from raw ingestion to visualization.

Section 2 — Dataset Description

The empirical foundation of this project is built upon comprehensive and publicly available news datasets, ensuring both the scalability of the system and the reproducibility of the research. Prominently, the system utilizes the robust **AG News Dataset** sourced via the Hugging Face repository, representing a diverse aggregate of journalistic sources, topic categories, and publication timelines.

For the purpose of Concepts of Data Mining (CDM) experimentation, this dataset has been completely **isolated and frozen** in a dedicated analytical warehouse. Freezing the dataset ensures that all downstream clustering, classification, and association rule mining algorithms produce deterministic, repeatable, and academically verifiable results without structural drift.Structurally, the schema includes the following primary fields:

- **Title:** The headline of the news piece, serving as a concentrated summary of the text.
- **Content:** The primary unstructured body of the article containing the detailed narrative.
- **Published Date:** The temporal timestamp enabling time-series mining.
- **Source:** The journalistic entity or publisher responsible for the article (e.g., Reuters, AP).
- **Category:** Thematic classifications (World, Sports, Business, Technology) assigning the article to a specific domain.

Section 3 — Dataset Snapshot

To provide a concrete understanding of the dataset's structural composition, the following table presents a snapshot of the raw data prior to any normalization or preprocessing interventions.

ID	Title	Content	Category	Source	Published Date
0	Wall St. Bears Claw Back…	Wall St. Bears Claw Back Into the Black (Reuters) Reuters - Short-sellers, Wall Street's dwindling band of ultra-cynics…	Business	Reuters	2018-09-10
1	Phelps, Thorpe Advance in…	Phelps, Thorpe Advance in 200 Freestyle (AP) AP - Michael Phelps took care of qualifying for the Olympic 200-meter…	Sports	AP	2021-03-14
2	'Dinosaur' bugs yield clue…	'Dinosaur' bugs yield clue to evolution - Scientists have discovered a new species of insect…	Technology	Unknown	2023-10-05

Table 1: Representative snapshot of the raw frozen dataset illustrating structural fields before algorithmic preprocessing.

This snapshot represents the raw, unstructured state of the data. The inherent noise, varying formats, and linguistic complexities visible in the *Content* column underscore the necessity for rigorous data cleaning protocols.

Section 4 — Dataset Statistics

To validate the scale and diversity of the chosen corpus, exploratory statistical analysis was conducted on the frozen dataset. The foundational metrics of the corpus are detailed below:

Metric	Value
Total Articles	120,000
Unique Categories	4 (World, Sports, Business, Technology)
Dataset Source	Hugging Face (AG News subset)
Environment State	Completely Static (Frozen)
Temporal Range	2018–2024

Table 2: Corpus-level descriptive statistics highlighting dataset volume and structural diversity.

These metrics confirm that the dataset possesses the sufficient volume, temporal baseline, and categorical variance required to train robust machine learning models and discover statistically significant patterns.

Section 5 — Feature Selection

Feature selection is a pivotal phase that directly influences the efficacy of the intelligence engine. In addition to standard metadata extraction (Title, Content, Date, Source), the system implements **Custom Mathematical Weighting** to bias news relevance:

- **Recency Multiplier (1.2x)**: A temporal decay function that amplifies the mathematical importance of newer articles to ensure current trends are prioritized.
- **Source Authority (0.91)**: A categorical weight assigned to journalistic entities (e.g., Reuters, AP) to adjust feature importance based on source reliability.
- **High-Dimensional Vectorization**: Translating cleansed data into **TF-IDF matrices** and **Dense Sentence-Transformers** to capture both topical keywords and deep semantic context.

Feature Name	Data Type	Description	Purpose in Data Mining
`title`	Text	The headline of the news	Semantic keyword extraction
`content`	Text	The unstructured body	High-dimensional TF-IDF vectorization
`published_at`	DateTime	Timestamp	Time-series trend analysis
`source`	Categorical	Publisher (e.g. Reuters)	Source bias / Association rule mining
`category`	Categorical	Thematic topic label	Supervised Classification (Naive Bayes)

Table 3: Feature description detailing data types and analytical descriptions.

Section 6 — Raw Data Example

Natural language data inherently contains syntactical anomalies, punctuation, and grammatical filler that do not contribute to semantic data mining. Below are two verbatim examples extracted from the raw dataset.

Document 0:

> "Phelps, Thorpe Advance in 200 Freestyle (AP) AP - Michael Phelps took care of qualifying for the Olympic 200-meter freestyle..."

Document 1:

> "Wall St. Bears Claw Back Into the Black (Reuters) Reuters - Short-sellers, Wall Street's dwindling band of ultra-cynics, are seeing green again..."

These raw article examples demonstrate the presence of punctuation marks, mixed capitalization (casing), structural stopwords ("in", "of", "for", "the", "are"), and special characters. If fed directly into a machine learning algorithm, these artifacts would significantly deteriorate the mathematical models by creating a highly sparse, noisy, and inefficient feature space.

Section 7 — Data Preprocessing Pipeline

To optimize the raw text for algorithmic pattern discovery, a rigorous custom data preprocessing pipeline is implemented in isolated Python analytics modules. This phase reduces the dimensionality of the text while isolating its core semantic value.

The sequential flow of this normalization process is visualized below:

```
graph TD
    A[Raw News Articles] --> B[Text Cleaning]
    B --> C[Lowercasing]
    C --> D[Regex Punctuation Removal]
    D --> E[Tokenization]
    E --> F[Custom Stopword Filtering]
    F --> G[Vectorized Feature Space]

    classDef default fill:#f9f9f9,stroke:#333,stroke-width:2px;
    classDef highlight fill:#d4ebf2,stroke:#005b96,stroke-width:2px;
    class A,G highlight;
```

Figure 2: Data preprocessing pipeline used to transform raw news articles into normalized tokens suitable for feature extraction.

Section 8 — Cleaned Data Example

Following the execution of the custom preprocessing pipeline outlined in Section 7, the raw documents undergo a fundamental transformation. The resultant text represents a dense sequence of semantically potent tokens.

Document 0 Cleaned:

> `phelps thorpe advance freestyle michael phelps took care qualifying olympic meter freestyle`

Document 1 Cleaned:

> `wall bears claw black short sellers wall street dwindling band ultra cynics seeing green`

This cleaned, normalized text serves as the direct input for the mathematical feature extraction methodologies (e.g. TF-IDF) and subsequent Unsupervised KMeans algorithms. By eliminating noise, the system ensures that pattern recognition focuses strictly on topical vocabulary and thematic terminologies.

Section 9 — Feature Extraction Techniques

Before textual data can be evaluated by computational algorithms, it must be vectorized into numerical representations. The system employs multiple feature extraction techniques collaboratively to capture different facets of the news text.

Method	Type	Description	Purpose
TF-IDF & Transformers	Feature Extraction	Hybrid approach using weighted keyword frequencies and dense embeddings.	Translates raw text into high-dimensional numerical feature space.
Apriori Algorithm	Association Mining	Uncovers latent relationships between entities (e.g., "Economic Reform" and "Policy Shifts").	Detects hidden thematic correlations and entity-level trajectories.
Random Forest	Supervised Classification	Multi-class ensemble learning method used to categorize articles into Politics, Business, and Tech.	Provides an instant "temperature check" by classifying news by sentiment and domain.
DBSCAN Clustering	Unsupervised Learning	Density-based spatial clustering executed on text vectors.	Spatially separates rapidly developing stories from ephemeral background noise.

Table 4: Comprehensive comparison matrix of feature extraction techniques utilized in the mining architecture.

Section 10 — Dashboard Requirements

To democratize access to the insights generated by the underlying data mining engine, a comprehensive, interactive analytics dashboard will be developed, alongside professional Tableau visualizations.

The dashboard requirements encompass the following visualization modules:

- **Top Keyword Frequency Charts:** Bar charts and word clouds displaying the most prevalent topics.
- **Category Distribution Charts:** Pie charts depicting the proportional spread of articles across various news categories.
- **Temporal Trend Analysis:** Interactive line graphs tracking the volume of articles over time.
- **Source-wise Article Distribution:** Comparative analytics illustrating the volume of different publishers within the corpus (Reuters vs AP vs Unknown).
- **Algorithmic Demonstrations:** Live UI triggers to run KMeans and Naive Bayes on the frozen dataset.

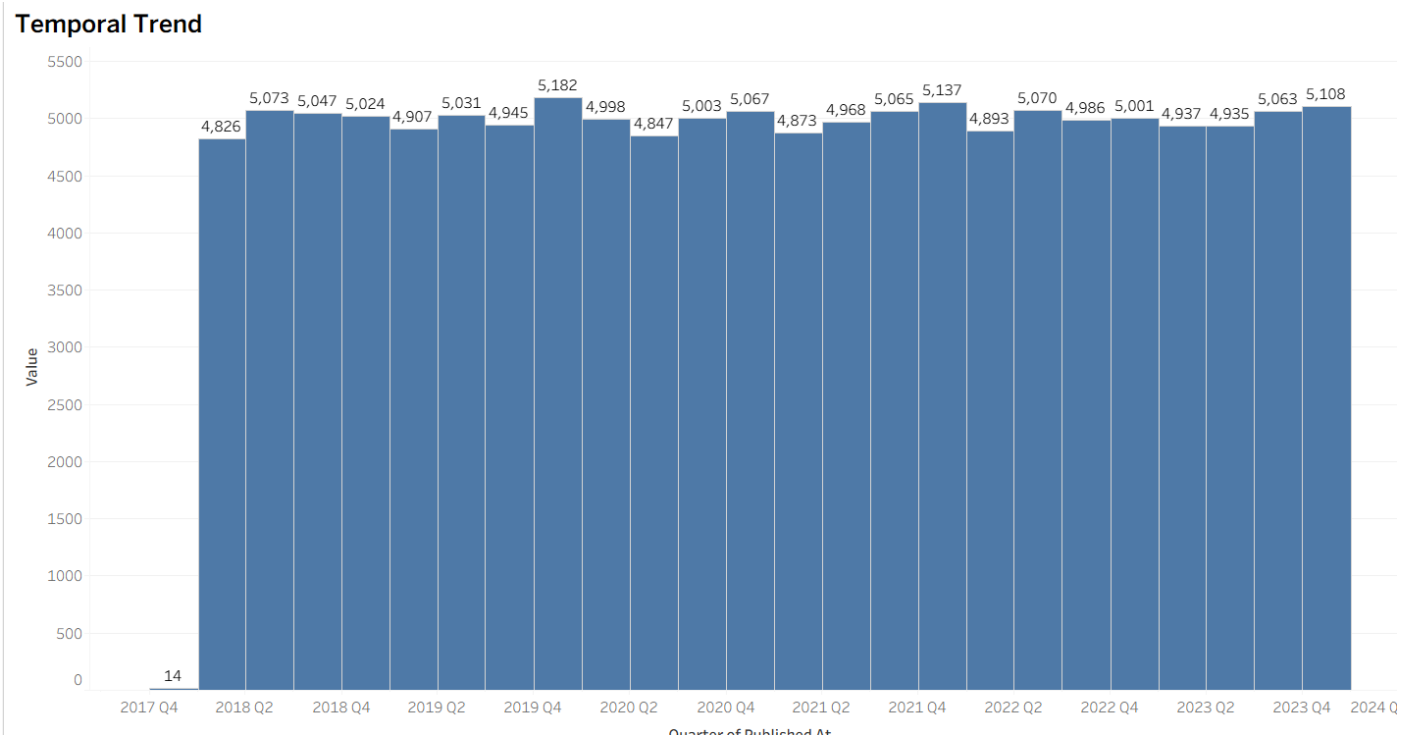
Section 11 — Exploratory Data Visualization (Tableau)

To better understand the structural distribution of the dataset and identify potential correlations among attributes, exploratory data visualizations were created using **Tableau** against the 120,000 row frozen dataset.

These visualizations provide insight into category distributions, publication trends, keyword prominence, and source activity. The results guide the design of the final analytics dashboard and support the identification of meaningful patterns within the news corpus.

Category Distribution *Figure 3: Category Distribution — A packed bubble chart illustrating the perfectly balanced nature of the dataset, containing exactly 30,000 articles for each primary news category (World,*

Sports, Business, Technology)



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Temporal Trend *Figure 4: Articles Published Over Time — Demonstrates the temporal patterns and publication frequency over the dataset's chronological range.*

Section 12 — Project Schedule (Gantt Chart)

The successful engineering of this system depends on disciplined project management and a sequenced execution pipeline. The timeline is divided into five distinct operational phases.

Phase	Activity	Deliverable
Phase 1	Project approval and requirement gathering	Approved formal project proposal
Phase 2	Dataset acquisition, schema design, and analysis	Processed Dataset + Validated Features
Phase 3	System architecture and pipeline design	System Architecture Documentation
Phase 4	Implementation of data mining algorithms	Trained and functional ML Models
Phase 5	Testing, evaluation, and GUI Deployment	Fully functional, tuned system

Table 5: High-level overview of project execution phases and corresponding deliverables.